

1. Purpose

This SOP ensures the safe handling, storage, transportation, and use of industrial gas cylinders to prevent accidents, leaks, or explosions.

2. Scope

Applicable to all personnel involved in using, storing, or transporting compressed gas cylinders in industrial, laboratory, and construction environments.

4. Types of Industrial Gases

Type Common Gases Risks

Flammable Gases Acetylene, Hydrogen, Propane Fire, explosion

Oxidizing Gases Oxygen, Nitrous Oxide Intensifies combustion

Inert Gases Nitrogen, Argon, Helium Asphyxiation

Toxic Gases Ammonia, Chlorine, Carbon Monoxide Poisoning, respiratory issues

Cryogenic Gases Liquid Nitrogen, Liquid Oxygen Frostbite, rapid expansion

5. Cylinder Identification & Labeling

Cylinders must have legible labels indicating the gas type.

Color coding varies by country—always follow local regulations.

NEVER use a cylinder if the label is missing or unreadable.

Cylinder Color Codes (Common for Reference)

Gas Cylinder Color

Oxygen Black/Green

Acetylene Maroon/Red

Nitrogen Grey

Argon Blue

Carbon Dioxide Silver/Green

Hydrogen Red

Helium Brown

6. Handling Procedures

✓ Do's:

✓ Always wear PPE (gloves, safety glasses, and steel-toe shoes).

✓ Inspect cylinders before use for leaks, dents, or corrosion.

✓ Use a cylinder cart for transport—never drag or roll a cylinder.

✓ Keep cylinder valves closed when not in use and use protective caps.

✓ Use only approved regulators and fittings.

✗ Don'ts:

- ✗ Never drop, throw, or strike cylinders.
- ✗ Do not use oil or grease near oxygen cylinders (can cause explosions).
- ✗ Never lift a cylinder by its cap or valve.
- ✗ Do not mix flammable and oxidizing gases in storage or use.
- ✗ Never attempt to modify, repair, or tamper with cylinders.

7. Storage Guidelines

Store cylinders upright, secured with chains or straps.

Keep in a well-ventilated, cool, dry area away from direct sunlight and heat sources.

Separate gases:

Flammable gases (e.g., Acetylene) must be stored 20+ feet away from Oxygen or use a fire-rated barrier.

Toxic and cryogenic gases must be stored in designated areas with warning signs.

Store empty and full cylinders separately and label them clearly.

Ensure fire extinguishers are nearby.

8. Transportation Safety

Use a trolley/cart with safety chains to transport cylinders.

Ensure valve protection caps are in place.

Keep cylinders upright and secured during transport.

DO NOT transport cylinders in enclosed vehicles (risk of gas buildup and explosion).

9. Usage & Operation

Before Use:

- ✓ Inspect for leaks using soapy water (never use flames).
- ✓ Ensure regulators and fittings match the gas type—never force connections.
- ✓ Open the valve slowly and fully using proper tools (no wrenches on hand-tight valves).

During Use:

- ✓ Keep flames and sparks away from flammable gases.
- ✓ Ventilate the area properly when using gases indoors.
- ✓ Close the cylinder valve when not in use—do not rely on the regulator alone.
- ✓ Do not over-pressurize hoses or equipment.

10. Emergency Procedures

Gas Leak:

Evacuate the area immediately.
Shut off valves if safe to do so.
Do NOT use electrical switches or flames.
Ventilate the area and inform safety personnel.

Fire involving gas cylinders:

Call emergency services immediately.
Evacuate and activate fire suppression systems.
Use Class B (flammable gases) or Class C (electrical) fire extinguishers.

Exposure to toxic gases:

Move affected person to fresh air.
Administer oxygen if available and seek medical attention.
Do NOT re-enter the area without protective equipment.

11. Inspections & Maintenance

Inspection Type	Frequency	Responsibility
Visual inspection for leaks, damage	Before each use	Operators
Cylinder integrity and label check	Monthly	Supervisors
Pressure test (hydrostatic test)	Every 3–5 years	Certified professionals

Cylinders with damage, leaks, or expired inspection dates must be removed from service immediately.

12. Regulatory Compliance

OSHA 29 CFR 1910.101 – Compressed Gas Safety

NFPA 55 – Compressed Gases & Cryogenic Fluids Code

DOT (49 CFR 173) – Cylinder Transport Regulations

13. Training & Documentation

Employees must receive annual training on gas cylinder handling, leak detection, and emergency response.

Maintain inspection records and logs for compliance audits